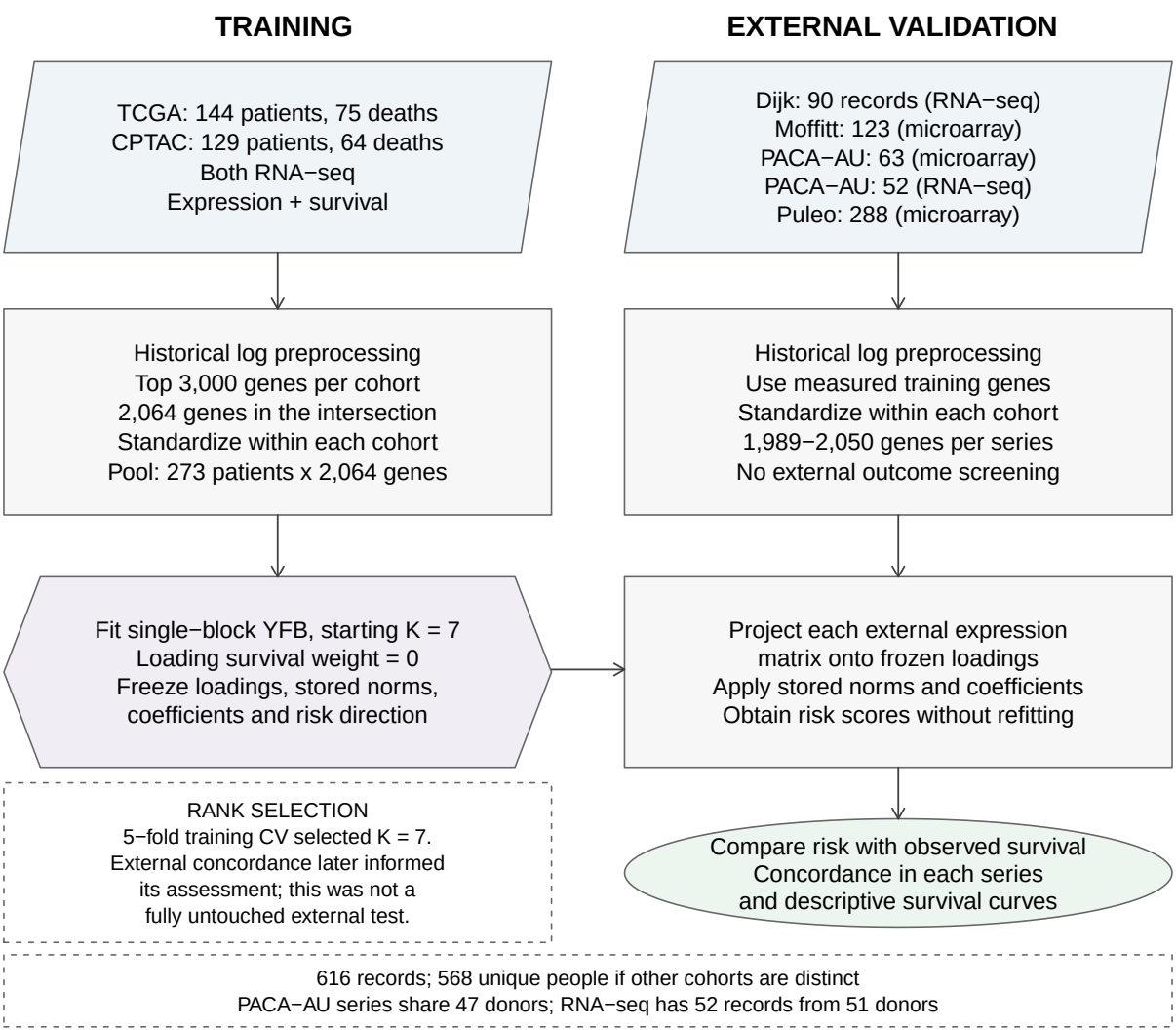
TRAINING
EXTERNAL VALIDATION
TCGA: 144 patients, 75 deaths
CPTAC: 129 patients, 64 deaths
Both RNA−seq
Expression + survival
Dijk: 90 records (RNA−seq)
Moffitt: 123 (microarray)
PACA−AU: 63 (microarray)
PACA−AU: 52 (RNA−seq)
Puleo: 288 (microarray)
Historical log preprocessing
Top 3,000 genes per cohort
2,064 genes in the intersection
Standardize within each cohort
Pool: 273 patients x 2,064 genes
Historical log preprocessing
Use measured training genes
Standardize within each cohort
1,989−2,050 genes per series
No external outcome screening
Fit single−block YFB, starting K = 7
Loading survival weight = 0
Freeze loadings, stored norms,
coefficients and risk direction
Project each external expression
matrix onto frozen loadings
Apply stored norms and coefficients
Obtain risk scores without refitting
RANK SELECTION
5−fold training CV selected K = 7.
External concordance later informed
its assessment; this was not a
fully untouched external test.
Compare risk with observed survival
Concordance in each series
and descriptive survival curves
616 records; 568 unique people if other cohorts are distinct
PACA−AU series share 47 donors; RNA−seq has 52 records from 51 donors